10/12/24 MIS

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[23ESCE32]

THIRD SEMESTER B. TECH EXAMINATION, DECEMBER - 2024

COMPUTER ORGANIZATION AND ARCHITECTURE

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**5 x2=10**
1.	**a.**	Compare and contrast Complex Instruction Set Computer (CISC) and Reduced Instruction Set Computer (RISC).	**L2**	**CO1**	**02**
	b.	Explain the operations of the control unit. Give one application for each of the main parts of computers.	**L2**	**CO1**	**02**
	c.	Differentiate Computer Architecture and Computer Organization.	**L2**	**CO1**	**02**
	d.	Explain various types of operations required for instructions.	**L2**	**CO2**	**02**
	e.	Define the terms CPI and MIPS	**L2**	**CO1**	**02**
		PART-B			**6 X 5=30**
2.	**a.**	Compare and contrast Multiprocessors and multi-computers. Mention 3 applications of each.	**L2**	**CO1**	**05**
		OR			
	b.	Calculate the 2's complement subtraction of a smaller number (101011) from a larger number (111001) or perform the arithmetic operation in binary numbers with negative numbers in signed 2's Complement form (-35) + (-40).	**L3**	**CO2**	**05**
3.	**a.**	Find the total time required to execute a program with 1000 instructions, where 25% of the instructions require 4 clock cycles, 40% require 5 clock cycles, and the remaining require 3 clock cycles, on a 1 GHz machine.	**L3**	**CO2**	**05**
		OR			
	b.	Determine IEEE standard for floating point format. Represent $(-15.625)_{10}$ in 32-bit IEEE format.	**L3**	**CO4**	**05**
4.	**a.**	Outline the schematic diagram for the Memory Read & Memory Write operation.	**L2**	**CO2**	**05**
		OR			
	b.	Discuss Addressing Modes. Mention different types of addressing Modes.	**L2**	**CO3**	**05**

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5.	a.	Design a common bus structure for the basic registers in the computing system.	L4	CO3	05
		OR			
	b.	Explain different types of addressing modes with examples.	L2	CO3	05
6.	a.	Describe different types of addressing instruction formats with examples.	L2	CO3	05
		OR			
	b.	Explain in detail about secondary storage devices with a neat diagram.	L2	CO5	05
7.	a.	Discuss the design steps that you will follow when you design a new CPU architecture using CPUSIM.	L2	CO5	05
		OR			
	b.	Illustrate the pipeline process in Pentium 4 and explain.	L2	CO4	05
		PART-C			**6 X 10=60**
8.	a.	Explain the connection between the processor and the memory with a neat diagram or describe the functional units of a computer with a neat diagram.	L2	CO2	10
		OR			
	b.	Describe how you'll evaluate a computer's performance. What are the criteria used to evaluate a computer's performance? Explain the basic Performance equation.	L2	CO6	10
9.	a.	Explain the Booth multiplication algorithm. Draw the flowchart and explain with an example.	L2	CO6	10
		OR			
	b.	Analyse and explain single bus structure with a neat diagram.	L4	CO3	10
10.	a.	A digital computer has a common bus system for 8 registers of 8 bits each. The bus is constructed with multiplexers. **i.** How many selection inputs are there in each multiplexer? **ii.** What is the size of each multiplexer? **iii.** How many multiplexers are there on the bus? **iv.** Write the hardware implementation of the above logic circuit.	L3	CO4	10

		OR			
	b.	Compare between memory-mapped I/O and I/O mapped I/O.	**L4**	**CO4**	**10**
11.	**a.**	Explain the carry save multiplier with neat sketch.	**L2**	**CO6**	**10**
		OR			
	b.	List the different types of interrupts and explain.	**L1**	**CO1**	**10**
12.	**a.**	Explain the memory system in the Pentium 4 processor.	**L2**	**CO2**	**10**
		OR			
	b.	Illustrate the CPU cache memory in the Athlon processor.	**L2**	**CO2**	**10**
13.	**a.**	Explain the general architecture of the Athlon processor.	**L2**	**CO1**	**10**
		OR			
	b.	Explain how DMA transfer is accomplished with a neat diagram.	**L2**	**C02**	**10**